

Henry Russell

Advanced Software Engineer

Work History

Advanced Software Engineer 2023 – Present

Zühlke, London

Barclaycard Payments – Marketing Site

Next.js, React, TypeScript, Contentful, Salesforce, Pricing API

Working for a large UK-based bank creating its marketing site. One of two developers on the project; helped set up and define the Contentful content model, built reusable templates with content editors, and integrated Salesforce and pricing API flows.

Co-op Funeralcare (Digital Products Team)

Vue, TypeScript, Python, Django, AWS, Contentful, Adobe Target

Built Vue microfrontends and Contentful/Jinja templates for customer journeys including Direct Cremation and funeral home finder. Delivered a finder app during a code-freeze incident, created a shared component library with web component support, implemented Adobe Target A/B tests, and contributed to Python/Django services, AWS integrations and monitoring.

PostFinance – AWS Bedrock Chatbot UI

Vue, TypeScript, AWS Bedrock

Built a lightweight Vue front end for an AWS Bedrock mortgage chatbot in a 10-day delivery window, focusing on clean architecture and client extensibility.

Professional Software Engineer 2022 – 2023

Zühlke, London

Handelsbanken (Mortgage Application Platform)

React, TypeScript, React Testing Library, Java, TDD, SAFE

Developed regulated mortgage application journeys under strict constraints, including custom routing, state management, validation and debounced autosave. Practised TDD and pair programming, collaborated closely with QA, and integrated with internal banking systems for document generation and data services.

Computer Science Teacher 2022 – 2024

Academic Tutors Surrey

One-to-one A-level Computer Science tuition, including lesson planning, explaining core topics and providing coursework feedback.

Software Engineer Intern Summer 2021

Royal Holloway

Lead developer for a student module-selection web platform integrating with internal university systems and used by students.

Computer Science Teaching Assistant 2020 – 2021

Royal Holloway

Supported first-year programming labs and assisted with marking coursework.

Student Ambassador 2019 – 2020

Represented the university at open days, campus tours and events within the Computer Science department.

Maths and English Tutor 2017 – 2019

Kumon, Frimley

Tutored students up to A-level, helping them progress through the Kumon programme and build confidence in core subjects.

Education

MSc Software Systems Engineering

University College London (UCL) · 2021 – 2022 · Distinction

Research collaboration with Intel and Microsoft: developed a Visual Studio plugin for gesture-based, touchless interaction with the IDE.

BSc Computer Science (Software Engineering)

Royal Holloway, University of London · 2018 – 2021 · First-Class Honours (avg 80%)

- Computer Science Prize for outstanding final-year project (2021).
- Best Second-Year Software Engineering Group Project (2020).
- Top academic performer (2019).

A Levels & GCSEs

- A Levels: Computer Science (A), EPQ (A), Psychology (A), Maths (B).
- 14 GCSEs A*–B, including A in Electronics and A grades in Computing, Maths, English and Sciences.

Skills and Projects

Front-End Engineering:

- React, Vue, Next.js, TypeScript, Contentful, component-driven architecture and modern UI patterns.
- Production marketing and customer-journey templates built with content editors, reusable components and testing in mind.
- **Portfolio website:** SEO-optimised portfolio for an actress with CI/CD deployment.
- **VTuber creator website:** interactive mini-games and old-web themed design.
- React interview practice lab with Vite, Playwright checks and GitHub Pages deployment guidance.

Full-Stack & Systems:

- **Yomu:** TypeScript userscript and web app for Japanese reading across websites, manga, PDFs and video, with popup dictionary, OCR, subtitle mining, JPDB, Anki, Jiten, Yomitan and a PWA study surface.
- Kanji learning automation tools which add AI-generated mnemonics.
- Voiceworks Toolkit for Japanese immersion workflows with local transcription, translation and semantic search.
- React dashboards backed by Python API services.
- **Royal Holloway module-selection system:** used internally by students.

Java:

- JavaFX chess engine with SQLite persistence layer.
- Autonomous Lego Mindstorms robot for search-and-retrieve tasks.
- Experience applying MVC, Observer and Factory patterns in test-driven builds.

Cloud, Tooling & Automation:

- AWS Solutions Architect Associate (2024).
- AWS, Azure, Docker, Linux workflows, CI/CD and automation.
- **UserScript Compiler:** converts userscripts into installable Manifest V3 extensions.
- **Templafy folder downloader:** retrieves original assets from library folders.
- **ChatGPT Automation Pro:** browser automation for repeatable ChatGPT workflows.
- Microservices-based multiplayer and real-time chess tournament application.

Hobbies and Interests

Web development, Japanese language study, chess and strategy games, psychology, photography, AI-augmented development, and lifelong learning.